

Does Financial Instruction Tuning Help LLMs?

Replicating the sentiment-classification slice of
Open-FinLLMs (Huang et al., 2024)

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Recap — the claim we're testing

Three weeks since Part A — quick reminder before we dive into results.

1



The paper

Open-FinLLMs (Huang et al., 2024) claims that financial instruction tuning materially improves downstream financial NLP tasks.



2



Part A (3 weeks ago)

Literature review, framing of the research question, and pre-registration of the experiment design.



3



Part B (today)

Subset replication on sentiment classification: does the financial tuning actually beat strong general LLMs?



Part B tests **one slice of the paper's claim** — sentiment classification on **FPB** and **FiQA-SA** — using current **7–8B models** and **classical baselines** as comparators.

The question

Open-FinLLMs (Huang et al., 2024) claims financial instruction tuning improves financial NLP.



Does TheFinAI's financial instruction tuning beat the LLaMA-family base it was built on?

Specifically, on sentiment classification — and how does it compare to classical baselines (FinBERT, VADER) and other 7–8B general-purpose LLMs (Mistral, Qwen2.5)?



Reproduced

FPB / FiQA-SA sentiment task;
4-bit inference of an 8B
financial LLM vs. 7–8B general
LLMs vs. classical baselines;
F1-macro / accuracy reporting.



Adapted

Substituted plutus-8B-instruct
(TheFinAI's 2025 successor) when
FinLLaMA-instruct was unpublished;
prompt templates A/B specifically
chosen to measure sensitivity.



Simplified

Inference only (no fine-tuning,
no multimodal, no trading sim).
300-sample subsample per dataset
for the LLM matrix to fit Modal T4
budget. Two prompts, two shot counts.

Reproducibility note

What changed since the paper, and what we did about it.



The model in the paper, TheFinAI/FinLLaMA-instruct, has been unpublished by the authors.



Substitute

TheFinAI/plutus-8B-instruct (Feb 2025) — same group, same LLaMA-3 8B base, same financial-tuning goal.



Finding in itself

LLM benchmarks become fragile when referenced models disappear.



Documented in our report

Our setup remains reproducible if the original model becomes available again.



Paper model



**TheFinAI/
FinLLaMA-instruct**
(Huang et al., 2024)

Status



Unpublished
by the authors

Our substitute



plutus-8B-instruct
(Feb 2025)

*Same group, same LLaMA-3 8B base,
same financial-tuning goal*

Method and experimental design

Pipeline, datasets, models, prompts, defenses against leakage.



Pipeline

Single unified sample schema; every downstream stage is dataset-agnostic.



Per-run dir — meta.json + predictions.jsonl + progress.json — checkpoint/resume built in



Modal Volume caches HF weights — multi-GB download paid once



Coverage tracked separately so parse failures aren't force-mapped to 'neutral'

Datasets

Two financial-sentiment benchmarks with different difficulty profiles.



Dataset	Source	Size	Labels	Note
Financial PhraseBank	Malo et al. 2014	690 test → 300 eval (sentences_75agree)	neg / neu / pos (3-class)	Annotator agreement $\geq 75\%$ — clean test signal
FiQA-SA	FiQA 2018	1,173 test → 300 eval	score $\in [-1, 1] \rightarrow$ 3-class with ± 0.10 band	Headlines + posts; noisier text, harder

- All models — LLMs and baselines — are evaluated on the same 300-sample subsample per dataset (identical ids, seed 42), so every confusion-matrix column sums to 300.
- Few-shot pool: only FPB train (FiQA has no train split) — preserves test purity
- Leakage caveat: both datasets predate every LLM here by 7-12 years and were almost certainly seen at pretraining time

Models

Five models across three families: financial-tuned, general 7–8B LLM, and classical.



Family	Model	Size / Type	Why included
 Financial-tuned	TheFinAI/plutus-8B-instruct	8B • LLaMA-3, 4-bit	Focal model — successor to FinLLaMA-instruct
 General-purpose LLM	mistralai/Mistral-7B-Instruct-v0.3	7B • 4-bit	Open LLaMA-family-like comparator
 General-purpose LLM	Qwen/Qwen2.5-7B-Instruct	7B • 4-bit	Strong open instruct baseline
 Classical specialised	ProsusAI/Finbert	110M • BERT classifier	Domain-specific small model — strong on FPB
 Classical lexicon	VADER	Rule-based	Cheap floor — what does no learning give you?

 All LLMs run at 4-bit on a single Modal T4 (16GB VRAM); LLaMA-3.1-8B-Instruct excluded — gated access not granted in time.

Two prompt templates (pre-registered)

Two templates is enough to measure prompt sensitivity — we don't pick the best, we report both.



Template A — minimalist

Classify the sentiment of the following financial text as positive, negative, or neutral.

Text: {text}

Sentiment:



Template B — structured + definitions

You are a financial analyst.

- Positive: favorable conditions, growth
- Negative: unfavorable, losses, risks
- Neutral: factual without clear implication

Text: {text}

Answer with one word only:

Experimental matrix and defenses

24 LLM runs ($3 \text{ models} \times 2 \text{ datasets} \times 2 \text{ templates} \times \{0, 3\}\text{-shot}$) · 4 baseline runs.



Deterministic decode (temperature=0) + seed=42 across every random draw



Few-shot pool drawn only from FPB train — never from test



Two templates pre-registered — both reported, neither picked post-hoc



Parse failures excluded from F1 and reported separately (no force map to 'neutral')



Reported metrics: *F1-macro (primary), accuracy, F1-weighted, per-class precision/recall, confusion matrix, parsing coverage, runtime.*

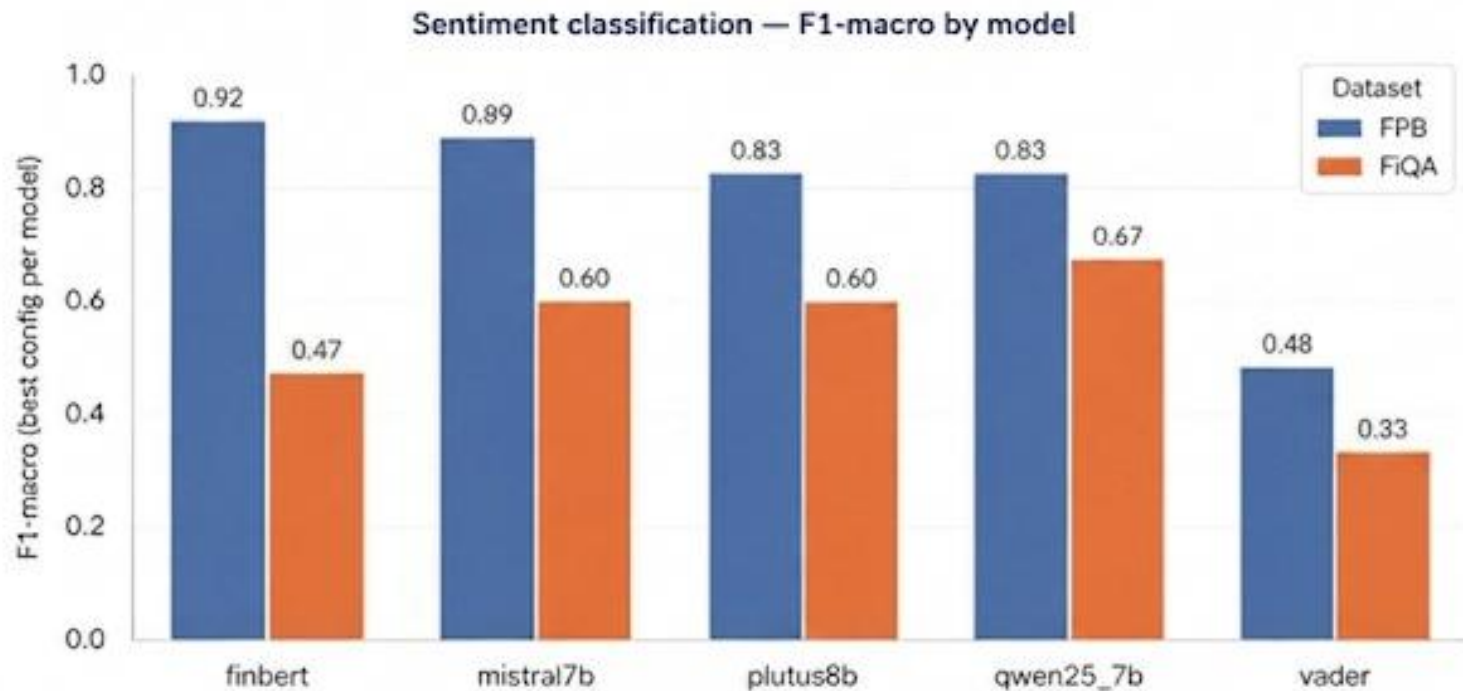
Results and error analysis

Headline numbers, where models break, and concrete examples.



Headline: F1-macro by model

Best configuration per model on each dataset. Bigger = better.



Key takeaways



FinBERT leads on FPB (0.92 F1-macro)



Qwen2.5-7B performs best on FiQA (0.67)



TheFinAI/Plutus-8B matches Mistral-7B on FPB but lags on FiQA



Rule-based VADER is the weakest across both datasets



Higher is better for F1-macro. Best per model across two templates and shots (0,3).

Headline numbers — best config per model × dataset

F1-macro is the primary metric (robust to class imbalance in FPB and FiQA).



Model	FPB · F1m	FPB · Acc	FiQA · F1m	FiQA · Acc
FinBERT (specialised, 110M)	0.917	0.927	0.469	0.473
Qwen2.5-7B-Instruct	0.832	0.860	0.673	0.717
Mistral-7B-Instruct-v0.3	0.890	0.903	0.599	0.620
plutus-8B-instruct (focal)	0.829	0.851	0.597	0.657
VADER (lexicon)	0.478	0.563	0.332	0.357



Key observations



FinBERT dominates FPB

Best F1-macro (0.925) and accuracy (0.935).



Qwen performs best on FiQA

Highest F1-macro (0.673) and accuracy (0.717).



Financial tuning does not consistently beat strong general LLMs

plutus is competitive but not best on either dataset.

 **Green** = per-dataset best. · **Best configs:** Mistral A/3-shot on FPB · Qwen B/0-shot on FiQA · plutus mid-pack.
Source: [results/summary/final_table.csv](#).

Three findings the numbers tell us



1. Financial instruction tuning did NOT pull ahead

plutus-8B (focal financial-tuned model) scored **0.829** F1m on FPB and **0.597** on FiQA.

Mistral-7B beat it on FPB (**0.890**) and Qwen2.5-7B beat it on FiQA (**0.673**).

On the paper's central claim — that financial instruction tuning improves downstream NLP — our subset replication says: **not clearly**, on these two datasets.



2. FinBERT dominates FPB; loses on FiQA — strong reverse on out-of-domain text

On FPB the 110M specialised classifier (**0.917** F1m) outperforms every 8B LLM.

On FiQA it drops to **0.469** — Qwen2.5-7B beats it by **20 points**.

Domain-trained classifiers crush LLMs in-domain but generalize poorly to noisier text.

A useful warning when picking models for production.



3. Prompt sensitivity is comparable to model choice

Mistral on FPB: Template A 0-shot **0.803** vs Template B 0-shot **0.683** — an 11-point gap from rewording alone.

Qwen on FiQA: Template B 0-shot **0.673** vs Template A 0-shot **0.578** — 9 points.

Plutus on FiQA: A 0-shot **0.597** vs B 0-shot **0.432** — 16 points.

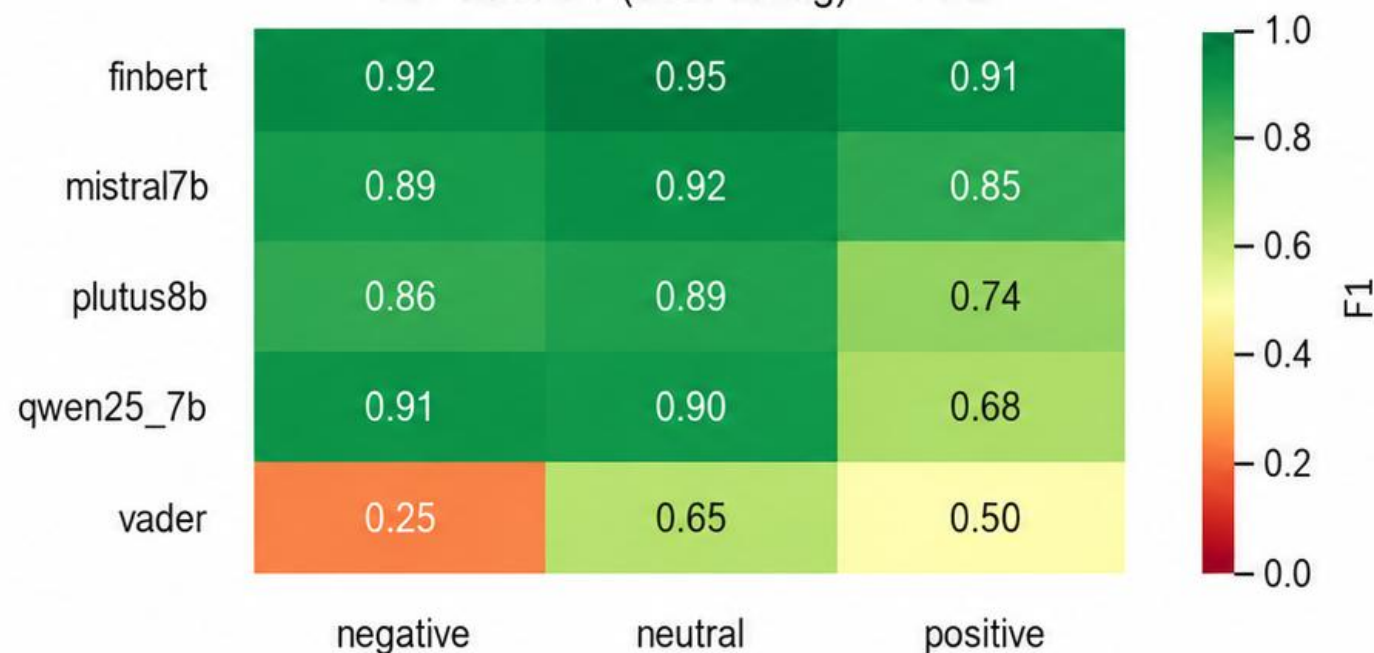
Single-prompt LLM benchmarks should not be trusted.

Per-class F1 — Financial PhraseBank

Class-level numbers expose hidden weaknesses an overall F1 can hide.



Per-class F1 (best config) — FPB



Key takeaways



Neutral is strongest

All LLMs score highest on neutral (0.89–0.95).



Positive is the weak spot

Biggest drop across models — especially for Plutus8B (0.74), Qwen2.5-7B (0.68) and VADER (0.50).



Negative is generally strong

Most LLMs are high on negative (0.86–0.92), except VADER (0.25).

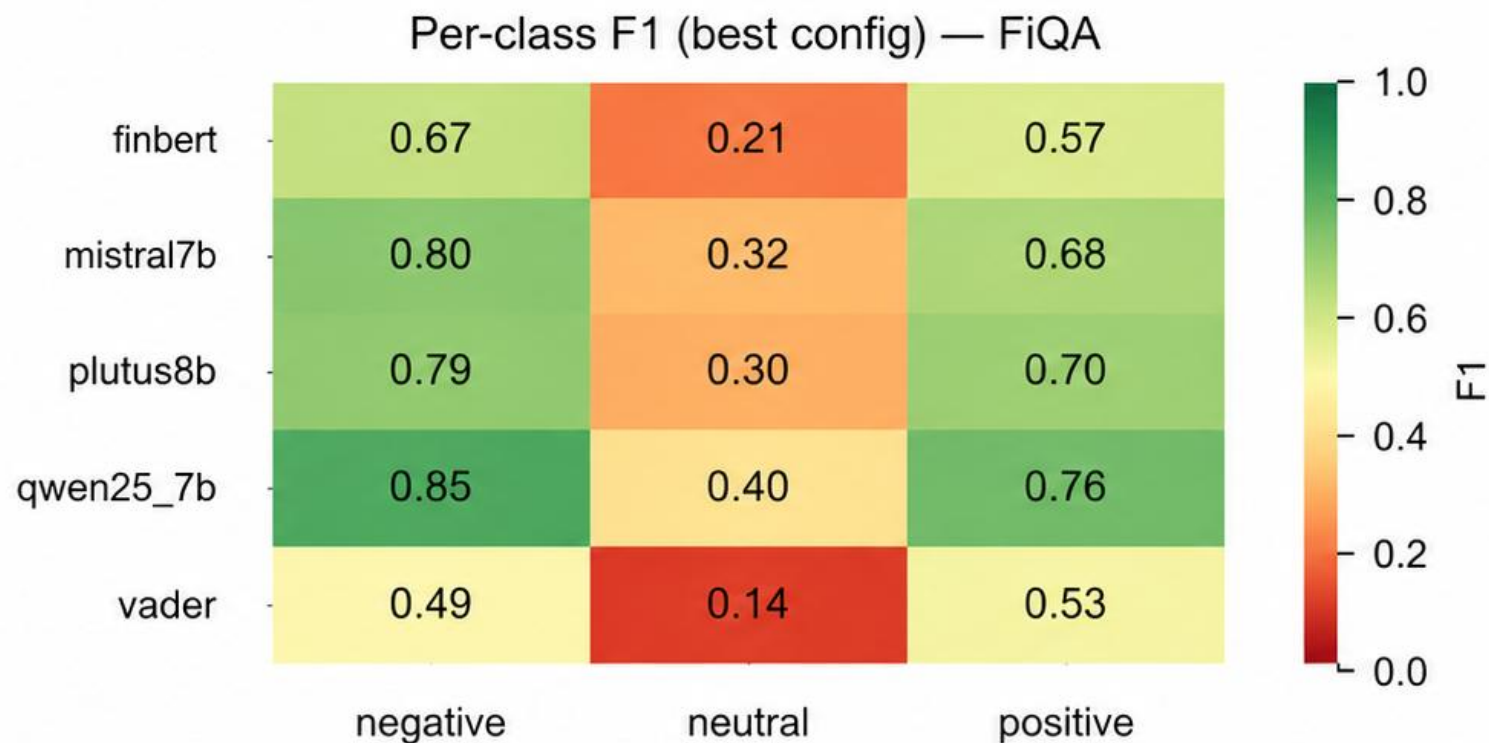


VADER lags significantly

Consistently weakest across all classes.

Per-class F1 — FiQA-SA

Note any class-specific gap between the financial model and the general LLMs.



Key takeaways



Neutral is the weakest class

All models struggle heavily on neutral sentiment.



Qwen performs best overall

Strongest across negative (0.85) and positive (0.76) classes.



Plutus stays competitive

Close to Mistral and strong on positive (0.70).



FinBERT collapses out-of-domain

Especially weak on neutral (0.21) FiQA examples.

Confusion matrices — best config per model × dataset

Rows: true label · Columns: predicted label · every cell now sums to 300 (matched subsample)

finbert · FPB F1m=0.92

true	neg	neu	pos
	45	0	0
	7	167	11
pos	2	2	66
	neg	neu	pos

predicted

finbert · FiQA F1m=0.47

true	neg	neu	pos
	47	39	6
	5	25	6
pos	16	86	70
	neg	neu	pos

predicted

mistral7b · FPB F1m=0.89

true	neg	neu	pos
	41	4	0
	5	172	8
pos	1	11	58
	neg	neu	pos

predicted

mistral7b · FiQA F1m=0.60

true	neg	neu	pos
	68	20	4
	6	25	5
pos	5	74	93
	neg	neu	pos

predicted

plutus8b · FPB F1m=0.83

true	neg	neu	pos
	43	2	0
	10	162	9
pos	2	21	46
	neg	neu	pos

predicted

plutus8b · FiQA F1m=0.60

true	neg	neu	pos
	86	4	2
	17	17	2
pos	22	56	94
	neg	neu	pos

predicted

qwen25_7b · FPB F1m=0.83

true	neg	neu	pos
	42	3	0
	2	176	7
pos	3	27	40
	neg	neu	pos

predicted

qwen25_7b · FiQA F1m=0.67

true	neg	neu	pos
	82	9	1
	8	25	3
pos	10	54	108
	neg	neu	pos

predicted

vader · FPB F1m=0.48

true	neg	neu	pos
	10	14	21
	13	110	62
pos	6	15	49
	neg	neu	pos

predicted

vader · FiQA F1m=0.33

true	neg	neu	pos
	32	33	27
	8	10	18
pos	23	84	65
	neg	neu	pos

predicted

Few-shot effect

Adding examples does not always improve performance

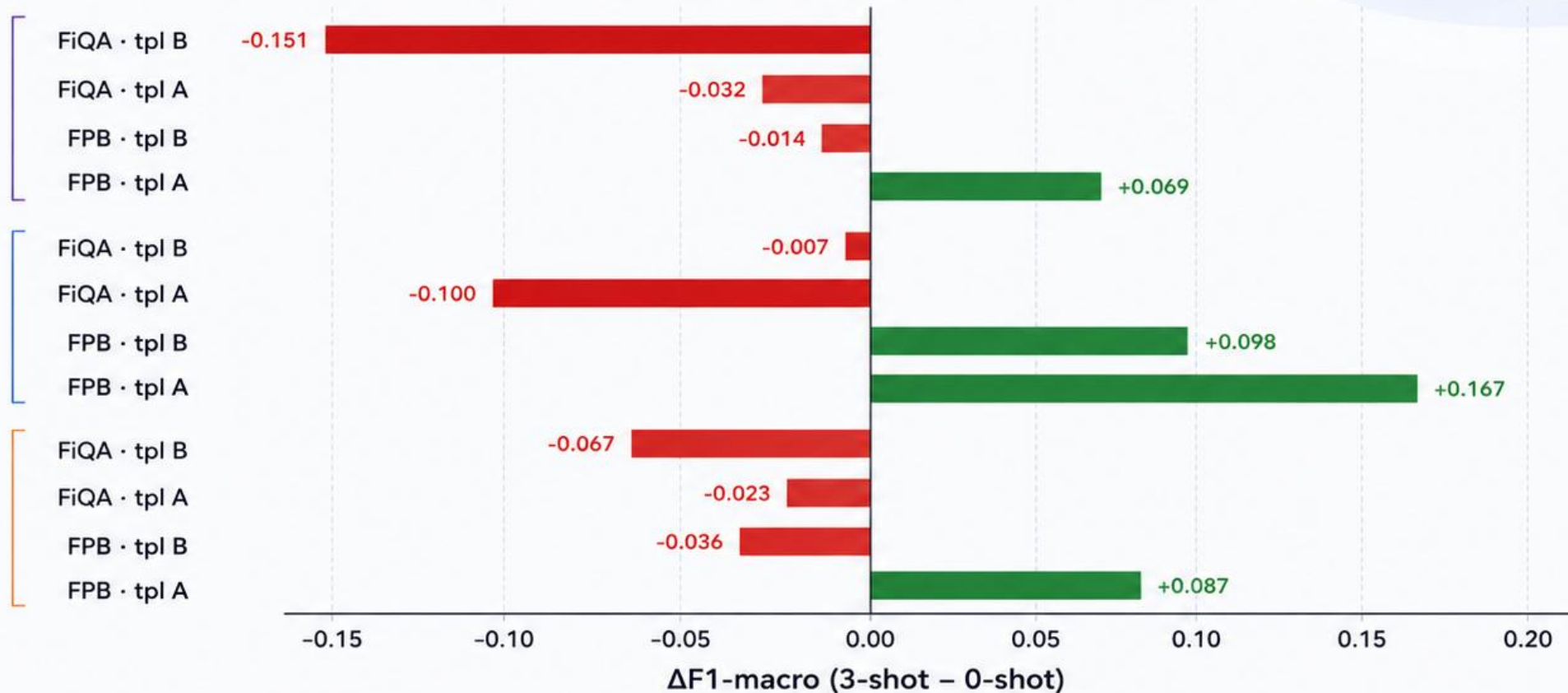



Qwen2.5-7B


Plutus-8B


Mistral-7B

Change in F1-macro when going from 0-shot to 3-shot ($\Delta F1\text{-macro}$)



Negative = performance decreased
3-shot hurt compared to 0-shot



Positive = performance improved
3-shot helped compared to 0-shot

Parsing coverage — does the model follow the format?

Coverage is reported separately so we don't paper over format failures.



LLM parsing coverage

	A-0	A-3	B-0	B-3
mistral7b-FPB	1.00	1.00	1.00	1.00
mistral7b-FiQA	1.00	1.00	1.00	1.00
plutus8b-FPB	1.00	0.98	1.00	1.00
plutus8b-FiQA	1.00	0.97	1.00	1.00
qwen25_7b-FPB	1.00	1.00	1.00	1.00
qwen25_7b-FiQA	1.00	1.00	1.00	1.00

template-shots

Concrete cases — focal-model errors on FPB

Real samples from results/summary/focal_error_sample.csv — each card = pattern + text + plutus prediction.



Numerical reasoning

"...net loss from continuing operations **narrowed** to 1.8 mln euro from 3.7 mln euro a year earlier."

Gold: positive · plutus-8B: negative · (FPB_00462)

Factual neutral → positive

"Timo Penttila has been **appointed** new manager responsible for the asset management of Nordea's institutional customers in Finland."

Gold: neutral · plutus-8B: positive · (FPB_01120)

Future-tense growth → wrongly neutral

"The company **plans to expand** into the international market through its subsidiaries and distributors from 2011 onwards."

Gold: positive · plutus-8B: neutral · (FPB_00380)

Negation / regulator hedge

"...the company **omitted** to tender for a substantial part of the works and as such they had rightfully been found non-responsive..."

Gold: neutral · plutus-8B: negative · (FPB_03174)

Where the system works

Strengths grounded in our numbers and confusion matrices.



- **FinBERT** on FPB → **0.925** F1m — best result in the matrix (in-domain specialist wins).



- **Mistral-7B** on **FPB** (A, 3-shot) → **0.890** F1m — general 7B closes most of the FinBERT gap with 3 examples.



- **Qwen2.5-7B** on **FiQA** (B, 0-shot) → **0.673** F1m — generalist beats FinBERT (0.482) out-of-domain.



- **≥97% parsing coverage** on all LLMs — no run poisoned by force-mapped failures.



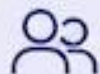
- **VADER** at **0.469 / 0.386** is the no-learning floor — every model clears it by **≥17** points.

Error categories — focal model misses (hand-categorized)

Hand-categorised plutus-8B errors on FPB (n=20 sample).



- **Numerical reasoning** — 'narrowed loss' / 'slight improvement' misread as negative



- **Factual neutrals → positive** — appointments, expansions, agreements
default upbeat



- **Negation** — 'did not miss expectations' flipped to negative



- **Domain jargon** — regulator commentary and hedged policy language
confuse the model



In our 20-case sample, plutus-8B was never uniquely correct where both Mistral and Qwen failed — consistent with the no-superiority conclusion.

A second test — does the claim hold on NER?

Open-FinLLMs claims financial tuning also wins named-entity recognition. We put that to the test.



The paper's claim — Open-FinLLMs reports that financial instruction tuning improves NER, beating general LLMs.



Why NER — An independent task. A genuine financial edge should generalize beyond sentiment, not vanish.



Dataset — FiNER-ORD — A financial NER benchmark with gold entity labels (PER / LOC / ORG).



Models & metric — plutus-8B (financial) vs Mistral-7B & Qwen2.5-7B (general) vs GLiNER-small / large (NER specialists). Strict span-level F1, 0-shot.



*Same research question as the sentiment slice — applied to a second, independent task.
If financial tuning is real, plutus-8B should lead here too.*

NER results — the claim did not reproduce

On strict span-level F1 the financial model lands last; general and specialist models lead.



Model	Family	Strict F1	Coverage
Qwen2.5-7B	General LLM	0.608	1.00
GLiNER-small	NER specialist	0.591	1.00
GLiNER-large	NER specialist	0.557	1.00
Mistral-7B	General LLM	0.519	1.00
plutus-8B	Financial LLM	0.372	0.835

Key observations

- plutus-8B is last — 0.37 strict F1 vs 0.61 for Qwen2.5-7B.
- It did not beat the general LLM base it was built on.
- A newer general model and modern NER specialists both win.
- Caveat: plutus coverage 0.835 — part of the gap is format-following, reported separately.

Bottom line: the paper's NER advantage did not reproduce — the substitute financial model lost to both general and specialist models.

Source: [results/summary_ner/final_table_ner.csv](#) · FiNER-ORD test set · strict span-level F1, 0-shot.

Two tasks, one answer

Sentiment and NER agree — the financial model's advantage did not show up.



Sentiment classification

0.829 / 0.597

F1-macro · FPB / FiQA-SA

plutus-8B: competitive, but never the best model.

Named-entity recognition

0.372

strict F1 · FiNER-ORD

plutus-8B: last of five models.

One task is an anecdote. Two independent tasks pointing the same way is a pattern.

Financial instruction tuning is narrow — it did not produce a generally stronger financial-NLP model.

Interactive analysis dashboard

Short live demo of the analysis interface used throughout the project.



1 Research-question verdict

2 F1 comparison chart

3 Prompt sensitivity view

At a glance

Runs in view

30

Best F1-macro

0.925

Best Accuracy

0.935

Mean coverage

99.84%

Does financial instruction-tuning help?

Compares the focal model (plutus-8B-instruct, the published successor to FinLLaMA-instruct) to the strongest non-financial 7-8B LLM on the same dataset.

FPB

✗ No — general LLM is better

Focal: `plutus-8B-instruct` (TheFinAI) (A, 3-shot) → F1m 0.829

Best general LLM: `Mistral-7B-Instruct-v0.3` (A, 3-shot) → F1m 0.890

Δ (focal – general): -0.061

FinBERT (specialised classifier): F1m 0.925

FiQA

✗ No — general LLM is better

Focal: `plutus-8B-instruct` (TheFinAI) (A, 0-shot) → F1m 0.597

Best general LLM: `Qwen2.5-7B-Instruct` (B, 0-shot) → F1m 0.673

Δ (focal – general): -0.076

FinBERT (specialised classifier): F1m 0.482

Per-example explorer + Highlights tab

Live demo: explore any example and instantly see where models disagree.



1



Pick any example

Open a sample (e.g., FPB_00462) and view all models' verdicts.

2



Compare model verdicts

See how each model classified the same example.

3



Open Highlights tab

Automatically surface and rank disagreement cases.

FPB_00462

Sample id: FPB_00462 · Gold: positive

Text: Finnish fibers and plastic products maker Suominen Corporation said its net loss from continuing operations narrowed to 1.8 mln euro (2.3 mln) in 2006 from 3.7 mln euro (4.8 mln) in 2005 .

	model	template	shots	pred	correct	parse_ok	raw
0	FinBERT-tone (analyst tone, 110M)	-	0	negative	☐	☒	Negative
1	FinBERT (ProsusAI, 110M)	-	0	positive	☒	☒	
2	Mistral-7B-Instruct-v0.3	A	0	neutral	☐	☒	Neutral The text
3	Mistral-7B-Instruct-v0.3	A	3	neutral	☐	☒	neutral Text: The
4	Mistral-7B-Instruct-v0.3	B	0	neutral	☐	☒	Neutral
5	Mistral-7B-Instruct-v0.3	B	3	neutral	☐	☒	neutral
6	plutus-8B-instruct (TheFinAI)	A	0	negative	☐	☒	negative
7	plutus-8B-instruct (TheFinAI)	A	3	negative	☐	☒	negative
8	plutus-8B-instruct (TheFinAI)	B	0	neutral	☐	☒	Neutral
9	plutus-8B-instruct (TheFinAI)	B	3	negative	☐	☒	negative

Models correct

1 / 15

Models wrong

14

Parse failures

0

Five lessons

What we learned from our evaluation across FPB and FiQA.



Financial tuning $\neq$ automatic win

plutus-8B was mid-pack; Mistral beat it on FPB, Qwen beat it on FiQA.



Prompt sensitivity $\approx$ model choice

9–16 F1 points moved by rewording alone — never trust a single-prompt benchmark.



Coverage matters

Force-mapping parse failures to 'neutral' would silently inflate accuracy.



Models can vanish

FinLLaMA-instruct was unpublished between paper and our run — real reproducibility risk.



Leakage caveat

FPB/FiQA predate the LLMs by 7–12 years — numbers are an upper bound.

What we'd do differently

Concrete next steps to improve the evaluation and make our results even more reliable.



- **Build a 2025 financial-news** held-out set to break leakage from FPB/FiQA pretraining



- **Run three or four prompt templates** instead of two — make sensitivity even more legible



- **Add an LLM-as-judge** sanity pass over a 100-sample slice — catch label noise



- **Try few-shot examples** drawn from the same dataset as the test (when available) — closer to real deployment



- **Include a smaller financial model** (e.g. finma-7b) and a non-financial 8B baseline of the same era for a tighter family comparison

What this means for finance

Stepping back from F1 numbers — what should a practitioner take away?



Is there a real edge to domain-specific financial LLMs?

In-domain, yes — FinBERT crushes FPB.

But an 8B financial-tuned LLM did NOT consistently beat 7B general LLMs in our matrix.



Does the edge justify the training + maintenance cost?

For a narrow, well-defined task — possibly.

For broad use — a strong general LLM + careful prompting looks like better ROI.



How critical are reproducibility and model availability?

Critical. The paper's model was unpublished mid-project.

In regulated finance (audit, compliance), an LLM that can disappear is a non-starter.



Final takeaway

Financial instruction tuning did not consistently outperform strong general LLMs.
Prompting strategy and dataset choice influenced performance just as much — and sometimes more — than the model itself.



plutus-8B vs strongest general models

FPB

0.829 vs 0.890

FiQA

0.597 vs 0.673



Prompt sensitivity

9 – 16 F1 points
(template A vs B)

Thank you

